

Bayes $\rightarrow$ Before:-

coin $\rightarrow$ $\frac{H}{T}$
 An email arrive check $\rightarrow$ spam / not spam

$\begin{pmatrix} 5R \\ 3B \\ 6-2 \end{pmatrix} \rightarrow 10$
 $P(R) = \frac{5}{10} = \frac{1}{2} = 0.5$
 $P(B) = \frac{3}{10}$
 $P(6) = \frac{2}{10} = \frac{1}{5} = 0.2$

A student wears glasses.
 C.S. def. $\rightarrow$ Conditional probability
 = probability after receiving new info.

Classroom = 100 students
 60B, 40H
 30B6, 10B6
 40
 $P(\text{stud. glasses}) = \frac{40}{100} = 0.4$
 S.S. = 60 Boys
 $P(\frac{\text{Glasses}}{\text{Boys}}) = \frac{30}{60} = 0.5$
 $\rightarrow$ Conditional probability reduces the sample space.
 $P(\frac{A}{B}) = \frac{P(A \cap B)}{P(B)}$

Independent events
 $P(A \cap B) = P(A)P(B)$
 coin toss $\Rightarrow P(H) = 0.5$, dice roll $\Rightarrow P(6) = 0.5$
 $P(\text{HEAD} \cap \text{2 even}) = 0.5 \times 0.5 = 0.25$

Naive Bayes $\rightarrow$ features are independent
 $P(A), P(B), P(A \cap B)$
 100 employees
 40M $\rightarrow$ 20 M coffee
 60F $\rightarrow$ 20 F coffee
 $P(\text{Male}) = \frac{40}{100} = 0.4$
 $P(\text{coffee}) = \frac{50}{100} = 0.5$
 $P(\text{M} \cap \text{coffee}) = \frac{20}{100} = 0.2$

Factory A $\rightarrow$ 70% products, Defective A $\rightarrow$ 2%
 Factory B $\rightarrow$ 30% products, Defect B $\rightarrow$ 5%
 What is the probability of a randomly selected product is defective?
 Defective product $\rightarrow$ Factory A / Factory B
 Defect from A $\rightarrow P(A) \cdot P(\text{Defect} | A)$
 $0.7 \times 0.02 = 0.014$
 Defect from B $\rightarrow 0.3 \times 0.05 = 0.015$
 $\frac{0.015}{0.014} \rightarrow 1.29 \times 10^{-1}$ answer
 $\frac{0.014}{0.029}$ less of total probability

$P(B) = \sum P(B|A_i)P(A_i)$
 Total probability =
 Sum of all possible routes leading to the events.

Medical test:-
 Disease prevalence $\rightarrow$ 1%
 Test accuracy $\rightarrow$ 99%
 A $\rightarrow$ positive
 Q. What is probability patient actually has the disease?

Disease prevalence $\rightarrow$ 1%
 100 $\rightarrow$ actually have disease.
 Test accuracy 99%
 2 $\rightarrow$ 100
 1 $\rightarrow$ 9900
 Test sensitivity = 99%
 99 $\rightarrow$ positive, 1 $\rightarrow$ ve

95% correctly identified
 9900
 $\rightarrow$ FP $\rightarrow$ 5%
 495 $\rightarrow$ FP

Bayes T.

$P(\text{Disease} | \text{Positive}) = \frac{P(\text{Positive} | \text{Disease}) P(\text{Disease})}{P(\text{Positive} | \text{Disease}) P(\text{Disease}) + P(\text{Positive} | \text{Healthy}) P(\text{Healthy})}$

$P(\text{Positive} | \text{Healthy}) \rightarrow$ FPR: false positive rate

Positive Negative
 Disease 99 1
 Healthy 495 9405
 $99 + 495 = 594$
 $\frac{99}{594} = 16.6\%$
 false positive paradox